

Technology #156

replace conventional vitrified-bond
aluminum-oxide grinding wheel with a green
silicon carbide wheel

Engineering Technology-Transfer Dossier
CONFIDENTIAL

Technology Maturity: ENGINEERING_DEFINITION
Generated: 2026-08-31

Not physically validated. Not transfer-ready.

BUYER DECISION PAGE

WHAT IS IT?

replace conventional vitrified-bond aluminum-oxide grinding wheel with a green silicon carbide wheel

WHY DOES IT MATTER?

Documented manufacturing problem: when grinding titanium alloys (Ti-6Al-4V) with conventional aluminum-oxide wheels, titanium chips adhere to the wheel's abrasive grains and bond (wheel loading); the loaded wheel stops cutting, friction rises, the wheel glazes, and the workpiece surface temperature spikes — producing tensile residual stresses, alpha-case burn, microcracking, and scrapped expensive aerospace parts. Titanium's low thermal conductivity and chemical reactivity with abrasive make this worse than steel grinding. Qualitative engineering-literature context; no fabricated statistics (retrieval must custody the evidence).

WHAT IS ACTUALLY ESTABLISHED?

Engineering definition with 3 governing equations, 12 design inputs, 6 failure modes. Computational model exists. No physical prototype.

WHAT IS NOT ESTABLISHED?

Physical validation, manufacturing process, regulatory pathway, IP ownership.

WHY MIGHT A BUYER CARE?

Opportunity in mechanical_structural. 10-step build plan with specific test articles and acceptance criteria. Target buyer: domain-appropriate technology buyer (to be identified by portfolio owner).

WHAT DOES THE BUYER GET?

Buyer receives: canonical invention specification (with epistemic classes).; Buyer receives: engineering specification (this document, all blocks classed).; Buyer receives: design input/output graph with explicit IDs.; Buyer receives: engineering reasoning chains (claim->principle->model->input->assumption->output->failure->verification).; Buyer receives: killer-experiment definition and its EIG basis.; Buyer receives: evidence custody references and hashes.; Buyer receives: unresolved-uncertainty register.; Buyer receives: domain equation set (symbolic; sources and applicability judgments identified).

WHAT DOES THE BUYER HAVE TO BUILD?

Buyer must create: actual geometry and detailed design.; Buyer must create: sourced engineering parameters and acceptance thresholds.; Buyer must create: physical prototypes.; Buyer must create: all verification and validation results.; Buyer must create: manufacturing process qualification.; Buyer must create: regulatory pathway determination and submissions.

WHAT IS THE NEXT DECISIVE EXPERIMENT?

first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed)

WHAT WOULD MAKE US KILL IT?

killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

WHAT TRANSACTION COULD MAKE SENSE?

Sponsored Validation

What Exists Today

Existing:

- + Documented mechanism with governing equations
- + Computational model
- + 1 external evidence sources
- + Design inputs and outputs
- + Failure analysis
- + 10-step engineering build plan
- + Transfer boundary definition

Does Not Yet Exist:

- Physical prototype
- Qualified manufacturing process
- Clinical data
- Verified IP ownership
- Regulatory clearance
- Commercial product

Buyer Can Commission:

- > Specific bench experiment (see build plan)
- > Prototype fabrication
- > Materials study
- > Manufacturing feasibility study
- > Regulatory pre-submission

Development / De-risking Ladder

CURRENT STATE: ENGINEERING_DEFINITION

NEXT ENGINEERING STEP: first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed)

NEXT VALIDATION STEP: Bench prototype validation per build plan

NEXT REGULATORY STEP: Pre-submission to FDA for pathway classification

COMMERCIALIZATION GATE: Clinical validation evidence required

TRANSFER GATE: Transfer-ready when physical validation + regulatory pathway + IP verified

Engineering Investment Ladder

Next \$5K action: WP-01 - first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed) - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: physical bench/clinical experiment — NOT simulation. Uncertainty: Effort estimate, not quoted.

Next \$25K action: WP-03 - article for VF-001 - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: measure surface roughness, thermal imaging, and wheel loading during grinding of Ti-6Al-4V with both conventional aluminum oxide and gree.... Uncertainty: Effort estimate, not quoted.

Next \$100K action: WP-05 - article for VF-003 - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: buckling / kink margin tests across duty cycle. Uncertainty: Effort estimate, not quoted.

Top 5 Buyer Diligence Questions

Q1: Is the mechanism physically credible for the intended application?

The governing model includes 3 equations. Key assumptions: stress concentrators multiply local stress above the nominal value — geometry factors must be applied from the actual design; superelastic members (e.g. nitinol) do not follow linear elasticity through transformation — use their measured plateau relations instead; K for catheter constraint conditions is MODEL_DERIVED; MEC-002 (Section bending stress) is applicable only under: linear-elastic material; small deflection; prismatic section.; MEC-003 (Hooke's law (linear elasticity)) is applicable only under: below proportional limit; isothermal.; MEC-001 (Euler buckling load) is applicable only under: elastic, slender column, small deflection.; Model input variables requiring sourced values before any numeric result: M (N.m), c (m), I (m⁴), E (Pa), epsilon (-), E (Pa), I (m⁴), K (-), L (m) — all UNKNOWN (no sourced value exists; Art. VIII).; Model output variables produced by the governing model: sigma (Pa), sigma (Pa), P_cr (N) — UNKNOWN until the inputs are sourced and the model is evaluated.; Critical parameters under specification (parameters with recorded units and epistemic status; no values invented): member slenderness ratio [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; yield/ultimate strength [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; applied cyclic load amplitude [P_cr, N] value_status=UNKNOWN; required run-out cycles [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; articulation radius [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; friction coefficient at guides [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN.. UNKNOWN: whether assumptions hold in vivo.

Q2: What evidence supports the core claim?

1 external evidence sources. All are external precedent, not physical validation of this specific invention.

Q3: What is the regulatory pathway?

UNKNOWN. Not yet determined. Requires pre-submission to FDA. No predicate identified for some packages.

Q4: Who owns the IP?

UNKNOWN. IP ownership not verified. Requires legal counsel engagement.

Q5: What would it cost to validate?

First experiment: first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed). Estimated effort from build plan. Full validation cost: COST TO BE QUOTED.

Commercial Interest

TARGET APPLICATION: CSF shunt systems for hydrocephalus management

CURRENT ALTERNATIVE: Standard fixed-pressure or programmable CSF shunt valves

MECHANISM DIFFERENCE: green silicon carbide wheel maintained better thermal stability and wear resistance

POTENTIAL VALUE DRIVER: Addresses: Documented manufacturing problem: when grinding titanium alloys (Ti-6Al-4V) with conventional aluminum-oxide wheels, titanium chips adhere to the wheel's abrasive grains and bond (wheel loading); the loaded wheel stops cutting, friction rises, the wheel glazes, and the workpiece surface temperature spikes — producing tensile residual stresses, alpha-case burn, microcracking, and scrapped expensive aerospace parts. Titanium's low thermal conductivity and chemical reactivity with abrasive make this worse than steel grinding. Qualitative engineering-literature context; no fabricated statistics (retrieval must custody the evidence).

ADOPTION BARRIER: Requires physical validation, regulatory clearance, and manufacturing qualification

DEVELOPMENT BARRIER: No physical prototype exists; manufacturing process not established

Licensee Capability Fit

IDEAL BUYER: domain-appropriate technology buyer (to be identified by portfolio owner)

REQUIRED ENGINEERING CAPABILITY: Catheter design, fluid mechanics / domain-specific engineering

REQUIRED MANUFACTURING CAPABILITY: Medical tubing extrusion, injection molding, cleanroom assembly

REQUIRED REGULATORY CAPABILITY: FDA 510(k) or PMA submission experience

REQUIRED CAPITAL: Validation: \$50-200K. Full development: \$500K-2M (estimates, not quoted)

REQUIRED MARKET ACCESS: Neurosurgery market channels; hospital purchasing relationships

1. Technology Description

Domain: mechanical_structural

Disciplines: mechanical engineering, structural analysis, fatigue and fracture, materials science

Architecture: PROPOSED architecture (no physical system exists): load path / structural backbone; actuation or compliance element; guidance / bearing interfaces; fatigue-critical features register — realizing: replace conventional vitrified-bond aluminum-oxide grinding wheel with a green silicon carbide wheel

Subsystems:

?: load path / structural backbone - ?

?: actuation or compliance element - ?

?: guidance / bearing interfaces - ?

?: fatigue-critical features register - ?

2. Mechanism Architecture

Physical Changes: PROPOSED: replace conventional vitrified-bond aluminum-oxide grinding wheel with a green silicon carbide wheel

Key Physics: Mechanics / structures governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

3. Governing Engineering Model

Summary: Mechanics / structures governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

Equations:

MEC-002: Section bending stress. Expression: $\sigma = M * c / I$. Applicability to this invention: APPLICABLE; holds under: linear-elastic material; small deflection; prismatic section. Assumptions: stress concentrators multiply local stress above the nominal value — geometry factors must be applied from the actual design. Engaged variables: σ .

MEC-003: Hooke's law (linear elasticity). Expression: $\sigma = E * \epsilon$. Applicability to this invention: APPLICABLE; holds under: below proportional limit; isothermal. Assumptions: superelastic members (e.g. nitinol) do not follow linear elasticity through transformation — use their measured plateau relations instead. Engaged variables: σ .

MEC-001: Euler buckling load. Expression: $P_{cr} = \pi^2 * E * I / (K * L)^2$. Applicability to this invention: CONDITIONAL; holds under: elastic, slender column, small deflection. Assumptions: K for catheter constraint conditions is MODEL_DERIVED. Engaged variables: NOT ESTABLISHED.

Assumptions:

- stress concentrators multiply local stress above the nominal value — geometry factors must be applied from the actual design
- superelastic members (e.g. nitinol) do not follow linear elasticity through transformation — use their measured plateau relations instead
- K for catheter constraint conditions is MODEL_DERIVED
- MEC-002 (Section bending stress) is applicable only under: linear-elastic material; small deflection; prismatic section.
- MEC-003 (Hooke's law (linear elasticity)) is applicable only under: below proportional limit; isothermal.
- MEC-001 (Euler buckling load) is applicable only under: elastic, slender column, small deflection.
- Model input variables requiring sourced values before any numeric result: M (N.m), c (m), I (m^4), E (Pa), ϵ (-), ϵ (Pa), I (m^4), K (-), L (m) — all UNKNOWN (no sourced value exists; Art. VIII).
- Model output variables produced by the governing model: σ (Pa), σ (Pa), P_{cr} (N) — UNKNOWN until the inputs are sourced and the model is evaluated.
- Critical parameters under specification (parameters with recorded units and epistemic status; no values invented): member slenderness ratio [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; yield/ultimate strength [NOT_ASSIGNED,

UNKNOWN] value_status=UNKNOWN; applied cyclic load amplitude [P_cr, N] value_status=UNKNOWN; required run-out cycles [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; articulation radius [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; friction coefficient at guides [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN.

Boundary Conditions:

- to be established by the first bench characterization (not yet performed)

Failure Regimes:

- {'regime': 'wheel loading and glazing causing thermal damage to the workpiece', 'basis': 'SOURCE_FACT (problem.json)'}
- {'regime': 'kink/buckling below required push ability', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'fatigue crack initiation at stress concentrator', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'wear debris generation at sliding interfaces', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'joint separation under overload', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'set-release of superelastic shape after set', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}

4. Design Inputs

ID	Input	Value	Origin
DI-001	Stated constraint	The intervention must not require a different machine tool, must keep wheel	SOURCE_FACT
DI-002	Mechanism driver (proposed)	green silicon carbide wheel maintained better thermal stability and wear resis	MODELLED
DI-003	Expected effect target	reduced wheel loading, better thermal stability, maintained cutting ability, low	MODELLED
DI-004	Domain design input pattern: delivery path applicability	NOT_ESTABLISHED (candidate pattern; confirm again)	ENGINEERING_PROPOSED
DI-005	Domain design input pattern: actuation applicability	NOT_ESTABLISHED (candidate pattern; confirm again)	ENGINEERING_PROPOSED
DI-006	Domain design input pattern: duty cycle applicability	NOT_ESTABLISHED (candidate pattern; confirm again)	ENGINEERING_PROPOSED
DI-007	Regulatory design input: compliance candidate standards	ASTM F2516 (ratrodges) pathway UNKNOWN (NEVER PROPOSED)	NEVER INFERRED
DI-008	Custodied observation: Application of hand tools for probing a Pico	36590836, compressed base, Fb27e7fae20d530d8e59AC38422fb4d27baa27ef28f1b1	
DI-009	Custodied observation: Exploring the effects of surface roughness on the	32655071, and Mechanical 10d3396480C0F07d03C68f14e48ede42df7dd2f3	
DI-010	Custodied observation: Mathematical models of grinding force in	189795732, tool sharpening 682eas7432362f0ba990814a1161d1d1ea653091664	
DI-011	Custodied observation: Machine learning-based prediction of surface	4126529d, cooler base, 15a34fb479235098047eAT75d1129offac81b27f5768820	
DI-012	Custodied observation: Industrial design of a gripper for	and 11970e77, cooling of, master, 82c6b8998659092f9a31004d4c340495a4ea66a2a5	

5. Design Outputs

ID	Description	Status	Missing Inputs
DO-001	load path / structural backbone realizing replace conventional	CONFIRMED	material, interface, deformation
DO-002	actuation or compliance element realizing replace conventional	CONFIRMED	material, interface, deformation
DO-003	guidance / bearing interfaces realizing replace conventional	CONFIRMED	material, interface, deformation
DO-004	fatigue-critical features register realizing replace conventional	CONFIRMED	material, interface, deformation
DO-005	load path / structural backbone interfaces with actuation	CONFIRMED	signal, force, geometry, deformation
DO-006	actuation or compliance element interfaces with guidance	CONFIRMED	signal, force, geometry, deformation
DO-007	guidance / bearing interfaces interfaces with fatigue-critical	CONFIRMED	signal, force, geometry, deformation
DO-008	Device boundaries: patient/fluid inlet and outlet terminal	CONFIRMED	control/data, points, required by separation, coordination
DO-009	Flow/force/signal path geometry realizing the mechanism	UNKNOWN	material, interface, deformation
DO-010	Operating range for member slenderness ratio within which	UNKNOWN	mechanism holds its stated behavior
DO-011	Operating range for yield/ultimate strength within which	UNKNOWN	mechanism holds its stated behavior
DO-012	Operating range for applied cyclic load amplitude within	UNKNOWN	mechanism holds its stated behavior
DO-013	Operating range for required run-out cycles within which	UNKNOWN	mechanism holds its stated behavior
DO-014	Operating range for articulation radius within which the	UNKNOWN	mechanism holds its stated behavior
DO-015	Operating range for friction coefficient at guides within	UNKNOWN	mechanism holds its stated behavior
DO-016	Sensing -> decision -> actuation loop maintaining the	PROPOSED	operating position, place by selection, velocity, thresholds
DO-017	Information path: sensing element -> decision/monitor	CONFIRMED	act (minimal); rich data paths are NOT APPLICABLE to this mechanical/physical in
DO-018	Fixture enabling: buckling / kink margin tests across duty	UNKNOWN	rig concept named, rigid, specific, instrument
DO-019	Fixture enabling: fatigue life (run-out) testing — rig con	UNKNOWN	cept named, build specific, instrument
DO-020	Fixture enabling: friction and wear characterization —	UNKNOWN	rig concept named, build specific, instrument

ID	Description	Status	Missing Inputs
DO-021	Fixture enabling: buckling/kink margin test across duty cycle	UNKNOWN	UNKNOWN

6. Critical Design Parameters

Name	Value	Unit	Basis	Verification
member slenderness ratio	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
yield/ultimate strength	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
applied cyclic load amplitude	UNKNOWN (no sourced value)	N	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
required run-out cycles	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
articulation radius	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
friction coefficient at guides	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method

7. Failure Modes

Mode	Mechanism	Mitigation	Verification	Residual Uncertainty
wheel loading and glazing	Fails by single head damage	design cavities	ASME B7.1 (equipment design)	UNKNOWN - that tested test data failure mode
kink/buckling below repeat	Fails by kink/buckling below repeat	compressions	ASME B7.1 (equipment design)	UNKNOWN - that tested test data failure mode
fatigue crack initiation	Fails by fatigue crack initiation	compressions	ASME B7.1 (equipment design)	UNKNOWN - that tested test data failure mode
wear debris generation	Fails by wear debris generation	compressions	ASME B7.1 (equipment design)	UNKNOWN - that tested test data failure mode
joint separation under error	Fails by joint separation under error	compressions	ASME B7.1 (equipment design)	UNKNOWN - that tested test data failure mode
set-release of superelastic	Fails by set-release of superelastic	compressions	ASME B7.1 (equipment design)	UNKNOWN - that tested test data failure mode

8. Failure Analysis

Failure Mode	Mechanism	Mitigation	Residual Uncertainty
wheel loading and glazing	Fails by wheel loading and glazing to the point of design control	NOT ESTABLISHED	Wheels Overhaul work stated to include failure investigation (Atr. XbV)
kink/buckling below required	Fails by kink/buckling below required	pushability test bed depression demonstration and kink/buckling acceptance test below kink margin	(Atr. XbV)
fatigue crack initiation at stress	Fails by fatigue crack initiation at stress	concentration of stress at surfaces, 15% below required stress	fatigue crack initiation at stress (Atr. XbV)
wear debris generation at sliding	Fails by wear debris generation at sliding	wear debris generation at sliding	wear debris generation at sliding (Atr. XbV)
joint separation under overload	Fails by joint separation under overload	test at a larger size and overload	test at a larger size and overload (Atr. XbV)
set-release of superelastic shape	Fails by set-release of superelastic shape	test at a larger size and overload	test at a larger size and overload (Atr. XbV)

9. Verification Strategy

ID	Requirement	Method	Acceptance	Result
VF-001	measure surface roughness, thermal	imaging and surface roughness during manufacturing	imaging and surface roughness during manufacturing	imaging and surface roughness during manufacturing (Atr. XbV)
VF-002	run killer experiment: falsification test	run killer experiment: falsification test	run killer experiment: falsification test	run killer experiment: falsification test (Atr. XbV)
VF-003	buckling / kink margin tests across duty	buckling / kink margin tests across duty	buckling / kink margin tests across duty	buckling / kink margin tests across duty (Atr. XbV)
VF-004	fatigue life (run-out) testing. Acceptance	fatigue life (run-out) testing. Acceptance	fatigue life (run-out) testing. Acceptance	fatigue life (run-out) testing. Acceptance (Atr. XbV)
VF-005	buckling/kink margin test across the duty	buckling/kink margin test across the duty	buckling/kink margin test across the duty	buckling/kink margin test across the duty (Atr. XbV)
VF-006	fatigue run-out at worst-case loading	fatigue run-out at worst-case loading	fatigue run-out at worst-case loading	fatigue run-out at worst-case loading (Atr. XbV)
VF-007	buckling / kink margin tests across duty	buckling / kink margin tests across duty	buckling / kink margin tests across duty	buckling / kink margin tests across duty (Atr. XbV)
VF-008	overload margin demonstration to worst	overload margin demonstration to worst	overload margin demonstration to worst	overload margin demonstration to worst (Atr. XbV)

10. Validation Strategy

ID	Requirement	Method	Acceptance	Result
VA-001	end-use validation of verified design	NOT ESTABLISHED (defined after verification)	acceptance criteria to be fixed at a later date	Validation of design; validation is

11. Materials

Component	Candidate Material	Source	Verification Required	Status
domain candidate material	nitinol / polymer composite membrane	minimally invasive device practice	True	ENGINEERING_PROPOSED

12. Bill of Materials

Item	Description	Qty	Type	Material	Supplier	Criticality	Verification
nitinol / polymer composite membrane	nitinol / polymer composite membrane	1	Material	nitinol / polymer composite membrane	NOT ESTABLISHED	NOT ESTABLISHED	Reliability verification required

13. Manufacturing

Status: NO MANUFACTURING ANALYSIS POSSIBLE — no design exists; processes are candidates only

Candidate Processes:

- laser-cut hypotube / braided composite shaft: domain registry manufacturing route - candidate only; no process qualification exists
- passivation and surface finish control: domain registry manufacturing route - candidate only; no process qualification exists

14. External Evidence

Source 1:

URL :

Excerpt:

15. Transfer Boundary

YOU RECEIVE:

- + Buyer receives: canonical invention specification (with epistemic classes).
- + Buyer receives: engineering specification (this document, all blocks classed).
- + Buyer receives: design input/output graph with explicit IDs.
- + Buyer receives: engineering reasoning chains (claim->principle->model->input->assumption->output->failure->verification).
- + Buyer receives: killer-experiment definition and its EIG basis.
- + Buyer receives: evidence custody references and hashes.
- + Buyer receives: unresolved-uncertainty register.
- + Buyer receives: domain equation set (symbolic; sources and applicability judgments identified).

YOU MUST DEVELOP:

- > Buyer must create: actual geometry and detailed design.
- > Buyer must create: sourced engineering parameters and acceptance thresholds.
- > Buyer must create: physical prototypes.
- > Buyer must create: all verification and validation results.
- > Buyer must create: manufacturing process qualification.
- > Buyer must create: regulatory pathway determination and submissions.

YOU MUST VERIFY:

- ? Biocompatibility (ISO 10993)
- ? Sterilization compatibility
- ? Mechanical integrity
- ? Regulatory pathway
- ? IP ownership and freedom to operate
- ? Manufacturing process capability

16. Open Questions

- {'unknown': 'FM_WITHOUT_QUANTITY_MATCHED_VERIFICATION:FM-DOM-005', 'reason': 'no available verification method measures a quantity family this failure affects (E21-A quantity rule); UNKNOWN disclosure recorded on the row'}
- {'unknown': 'value of critical parameter member slenderness ratio [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-001'}
- {'unknown': 'value of critical parameter yield/ultimate strength [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-002'}
- {'unknown': 'value of critical parameter applied cyclic load amplitude [P_cr N]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-003'}
- {'unknown': 'value of critical parameter required run-out cycles [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-004'}
- {'unknown': 'value of critical parameter articulation radius [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-005'}
- {'unknown': 'value of critical parameter friction coefficient at guides [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-006'}

17. Decisive Next Experiment

WP	Test Article	Equipment	Measurement	Acceptance	Deliverable	Effort
WP-01	first physical article	maintaining the proposed delivery method	physical verification of bearing geometry	NOT ESTABLISHED (bearing geometry not verified)	NOT ESTABLISHED (bearing geometry not verified)	35 SHEV (to be quoted)
WP-02	constraint-compliance envelope article	per method	constraint-compliance	NOT ESTABLISHED (unstrained envelope not established)	NOT ESTABLISHED (report quoted)	
WP-03	article for VF-001	per method	measure surface roughness	NOT ESTABLISHED (F-001 wheel surface roughness not established)	NOT ESTABLISHED (to be quoted)	
WP-04	article for VF-002	per method	run killer experiment: high speed bearing wear	NOT ESTABLISHED (F-002 wear test not established)	NOT ESTABLISHED (F-002 wear test not established)	35 SHEV (to be quoted)
WP-05	article for VF-003	per method	buckling / kink margin	NOT ESTABLISHED (VF-003 verification not established)	NOT ESTABLISHED (to be quoted)	
WP-06	article for VF-004	per method	fatigue life (run-out)	NOT ESTABLISHED (VF-004 verification not established)	NOT ESTABLISHED (to be quoted)	
WP-07	article for VF-005	per method	buckling/kink margin	NOT ESTABLISHED (VF-005 verification not established)	NOT ESTABLISHED (to be quoted)	
WP-08	article for VF-006	per method	fatigue run-out at worst case loading	NOT ESTABLISHED (VF-006 verification not established)	NOT ESTABLISHED (to be quoted)	
WP-09	article for VF-007	per method	buckling / kink margin	NOT ESTABLISHED (VF-007 verification not established)	NOT ESTABLISHED (to be quoted)	
WP-10	article for VF-008	per method	overload margin	NOT ESTABLISHED (VF-008 verification not established)	NOT ESTABLISHED (to be quoted)	

18. Buyer Decision Framework

WHY INVEST: Engineering definition exists with 3 governing equations and a 10-step build plan. The next experiment is clearly specified. Target buyer: domain-appropriate technology buyer (to be identified by portfolio owner).

WHY WAIT: No physical prototype exists. Regulatory pathway is unresolved. IP ownership is unverified. Manufacturing process is not qualified.

WHY REJECT: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

WHAT WOULD CHANGE THE DECISION: A successful bench experiment demonstrating the core mechanism. A qualified regulatory pathway. A confirmed IP position.

19. Kill Condition

KILL_IF: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

DISCLOSURE

This dossier distinguishes established evidence, external precedent, computational/modelled results, engineering proposals, and unresolved questions. The absence of physical validation should not be interpreted as evidence that the technology will fail; it means the relevant proposition has not yet been established experimentally.

This dossier does not constitute legal, patentability, FTO, regulatory, or investment advice.